

**Topics:** 12.5 Equations of Lines and Planes (Part II); 12.6 Cylinders and Quadric Surfaces; 13.1 Vector Functions and Space Curves

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- Find parametric equations for the line of intersection of the planes

$$2x + 3y + 5z = 7 \quad \text{and} \quad x - y + 2z = 3.$$

Normal vectors to the planes are

$$\mathbf{n}_1 = \langle 2, 3, 5 \rangle, \quad \mathbf{n}_2 = \langle 1, -1, 2 \rangle.$$

A direction vector for the line of intersection is

$$\mathbf{v} = \mathbf{n}_1 \times \mathbf{n}_2 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 3 & 5 \\ 1 & -1 & 2 \end{vmatrix} = 11\mathbf{i} + \mathbf{j} - 5\mathbf{k} = \langle 11, 1, -5 \rangle.$$

To find a point on the line, we can set  $y = 0$  which gives the system of two equations:

$$2x + 5z = 7, \quad x + 2z = 3.$$

Solving this system, we find  $x = 1$  and  $z = 1$ . Thus  $P_0(1, 0, 1)$  is on the line. Therefore,

$$\langle x, y, z \rangle = \langle 1, 0, 1 \rangle + t\langle 11, 1, -5 \rangle$$

is an equation of the line, so parametric equations for the line are

$$x = 1 + 11t, \quad y = t, \quad z = 1 - 5t.$$

- Find an equation of the plane through  $P(7, -2, -4)$  that is parallel to the plane  $z = 4x - 5y$ .

Rewrite the equation of the given plane as  $4x - 5y - z = 0$ , so a normal vector is  $\mathbf{n} = \langle 4, -5, -1 \rangle$ .

A parallel plane through  $P(7, -2, -4)$  has the same normal vector:

$$4(x - 7) - 5(y + 2) - (z + 4) = 0.$$

This can be rewritten as:

$$4x - 5y - z = 42.$$

- Find an equation of the plane that passes through the point  $P(10, -1, 5)$  and contains the line with symmetric equations  $\frac{x}{4} = y + 6 = \frac{z}{5}$ .

Setting  $\frac{x}{4} = y + 6 = \frac{z}{5} = t$ , we get a parametric form of the line:

$$x = 4t, \quad y = t - 6, \quad z = 5t,$$

so one vector parallel to the plane is  $\mathbf{v}_1 = \langle 4, 1, 5 \rangle$ .

Taking  $t = 0$  gives a point on the line:  $A(0, -6, 0)$ .

The vector  $\overrightarrow{AP}$  is also parallel to the plane:

$$\mathbf{v}_2 = \overrightarrow{AP} = \langle 10 - 0, -1 - (-6), 5 - 0 \rangle = \langle 10, 5, 5 \rangle.$$

Therefore a normal vector to the plane is

$$\mathbf{n} = \mathbf{v}_1 \times \mathbf{v}_2 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & 1 & 5 \\ 10 & 5 & 5 \end{vmatrix} = \langle -20, 30, 10 \rangle.$$

Using point  $P(10, -1, 5)$ , the plane equation is

$$-20(x - 10) + 30(y + 1) + 10(z - 5) = 0.$$

This can be rewritten as:

$$-20x + 30y + 10z = -180.$$

4. Determine whether the line  $L$  given by  $\mathbf{r}(t) = \langle -2t, 2 + 7t, -1 - 4t \rangle$  intersects the plane given by  $4x + 9y - 2z + 8 = 0$ .

Write parametric equations for  $L$ :

$$x = -2t, \quad y = 2 + 7t, \quad z = -1 - 4t.$$

Substitute into the plane equation:

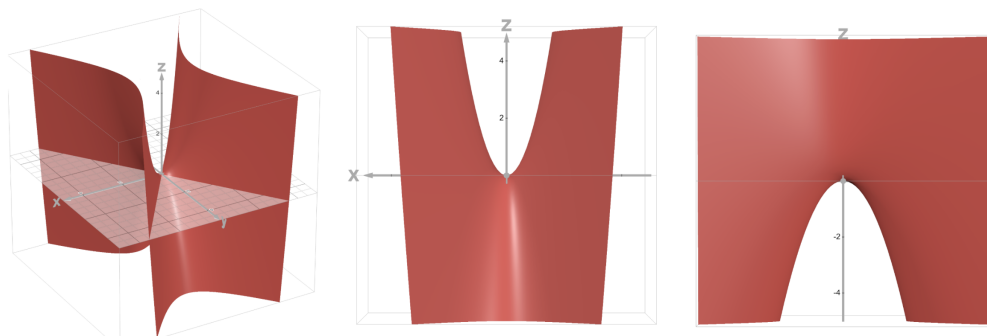
$$4(-2t) + 9(2 + 7t) - 2(-1 - 4t) + 8 = 0$$

$$-8t + 18 + 63t + 2 + 8t + 8 = 0 \implies 63t + 28 = 0 \implies t = -\frac{4}{9}.$$

Thus the line intersects the plane. The intersection point is

$$(x, y, z) = \left( -2 \left( -\frac{4}{9} \right), 2 + 7 \left( -\frac{4}{9} \right), -1 - 4 \left( -\frac{4}{9} \right) \right) = \left( \frac{8}{9}, -\frac{10}{9}, \frac{7}{9} \right).$$

5. A surface is shown below from three different viewpoints.



Which of the following equations represents the surface?

- A.  $x^2 + y^2 + z^2 = 1$
- B.  $z = 1 - x^2 - 2y^2$
- C.  $z^2 = x^2 + y^2$
- D.  $z = 2x^2 - y^2$

We observe from the graphs that the surface has a saddle shape: it curves upward in one direction and downward in a perpendicular direction. By process of elimination:

- $x^2 + y^2 + z^2 = 1$  is a sphere.
- $z = 1 - x^2 - 2y^2$  is an elliptic paraboloid opening downward (both  $x$ - and  $y$ -traces open down).
- $z^2 = x^2 + y^2$  is a circular cone (since  $z = \pm\sqrt{x^2 + y^2}$ ).

None of Choices A–C has a saddle shape. Thus the correct choice is D. We can check this by considering traces: if  $y = 0$ , then  $z = 2x^2$  opens up; and if  $x = 0$ , then  $z = -y^2$  opens down.

6. Find the domain of the vector function

$$\mathbf{r}(t) = \left\langle \ln(t+1), \frac{t}{4-t^2}, \sqrt{4-t} \right\rangle.$$

We first find the domain of each component of the vector function.

The domain of  $\ln(t+1)$  is given by  $t+1 > 0$ , so it is the set  $(-1, \infty)$ .

The domain of  $\frac{t}{4-t^2} = \frac{t}{(2+t)(2-t)}$  is given by  $t \neq 2$  and  $t \neq -2$ . In interval notation, that is

$$(-\infty, -2) \cup (-2, 2) \cup (2, \infty).$$

The domain of  $\sqrt{4-t}$  is given by  $4-t \geq 0$ , so it is the set  $(-\infty, 4]$ .

The domain of  $\mathbf{r}(t)$  is the intersection of the domains of its components. So the domain of  $\mathbf{r}(t)$  is

$$(-1, 2) \cup (2, 4].$$

7. Evaluate the limit  $\lim_{t \rightarrow 1} \mathbf{r}(t)$ , where

$$\mathbf{r}(t) = \left\langle \frac{t^2 - 1}{t^2 - 3t + 2}, \frac{t - 1}{\sqrt{t + 3} - 2}, \frac{\sin(t - 1)}{t - 1} \right\rangle.$$

To find the limit of a vector-valued function, we must compute

$$\lim_{t \rightarrow 1} \mathbf{r}(t) = \left\langle \lim_{t \rightarrow 1} \frac{t^2 - 1}{t^2 - 3t + 2}, \lim_{t \rightarrow 1} \frac{t - 1}{\sqrt{t + 3} - 2}, \lim_{t \rightarrow 1} \frac{\sin(t - 1)}{t - 1} \right\rangle.$$

For the first component,

$$\lim_{t \rightarrow 1} \frac{t^2 - 1}{t^2 - 3t + 2} = \lim_{t \rightarrow 1} \frac{(t + 1)(t - 1)}{(t - 1)(t - 2)} = \lim_{t \rightarrow 1} \frac{t + 1}{t - 2} = \frac{2}{-1} = -2.$$

For the second component, rationalize:

$$\lim_{t \rightarrow 1} \frac{t - 1}{\sqrt{t + 3} - 2} = \lim_{t \rightarrow 1} \frac{(t - 1)(\sqrt{t + 3} + 2)}{(t + 3) - 4} = \lim_{t \rightarrow 1} (\sqrt{t + 3} + 2) = \sqrt{4} + 2 = 4.$$

For the third component, use L'Hôpital's rule:

$$\lim_{t \rightarrow 1} \frac{\sin(t - 1)}{t - 1} = \lim_{t \rightarrow 1} \frac{\cos(t - 1)}{1} = \frac{\cos(0)}{1} = 1$$

Therefore,

$$\lim_{t \rightarrow 1} \mathbf{r}(t) = \langle -2, 4, 1 \rangle.$$

## SOME USEFUL DEFINITIONS, THEOREMS, AND NOTATION

**Equations of a plane.** The plane through  $(x_0, y_0, z_0)$  with normal vector  $\langle a, b, c \rangle$  is given by

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0.$$

By collecting terms, we can rewrite this equation in the form

$$ax + by + cz = d$$

where  $d = ax_0 + by_0 + cz_0$  is a constant. Two planes are parallel if their normal vectors are parallel, and they are perpendicular if their normal vectors are orthogonal.

**Vector functions.** A **vector-valued function** (or **vector function**) is a function whose domain is a set of real numbers and whose range is a set of vectors. In three dimensions, a vector-valued function is written as

$$\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k},$$

where  $f$ ,  $g$ , and  $h$  are real-valued **component functions**. The domain of  $\mathbf{r}(t)$  is the set of all values  $t$  for which  $f(t)$ ,  $g(t)$ , and  $h(t)$  are all defined. Limits of vector functions are defined component-wise: For a real number  $a$ ,

$$\lim_{t \rightarrow a} \mathbf{r}(t) = \left\langle \lim_{t \rightarrow a} f(t), \lim_{t \rightarrow a} g(t), \lim_{t \rightarrow a} h(t) \right\rangle,$$

provided the three component limits exist.

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### Suggested Textbook Problems

Section 12.5: 1-72

Section 12.6: 1-49

Section 13.1: 1-58